MZUMBE UNIVERSITY

FACULTY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTING SCIENCE STUDIES

A PROJECT REPORT

ON

E-COMMERCE PLATFORM FOR LOCAL ARTISANS TO SHOWCASE AND SELL HANDCRAFTED PRODUCTS

BY

FARIDI K. SULEIMANI

REG. NO: 14323007/T.22

PROGRAMME: ICTB-III

(BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY WITH BUSINESS)

SUPERVISED

BY

DR. MORICE

A Project Report Submitted to the Faculty of Science in Partial Fulfillment for the award of the Bachelor of Science in Information Technology with Business of

Mzumbe University

2025

DECLARATION

I, Faridi K. Suleimani, declare that this project proposal titled “E-commerce platform for local artisans to showcase and sell handcrafted products " is my original work, and it will be carried out in accordance with the regulations of the University of Mzumbe for partial fulfillment of the requirements for the degree award of Bachelor of Science in Information Technology and Systems.

SIGNED ……

DATE ……..…

SUPERVISOR’S SIGNATURE

….……

DATE ……

ACKNOWLEDGEMENT

I express my deepest gratitude to the Almighty God for granting me strength, wisdom, and perseverance throughout this journey. My sincere appreciation extends to my parents, whose unwavering love, encouragement, and sacrifices have been my greatest source of inspiration. I am also profoundly grateful to my relatives and friends for their continuous support and motivation.

A special note of thanks goes to Dr. Morice**,** my project supervisor, for his invaluable guidance, insightful feedback, and unwavering support, which have been instrumental in the successful completion of this work. His expertise and encouragement have truly shaped my understanding and execution of this project.

Lastly, I extend my heartfelt appreciation to my faculty members, classmates, and all those who, in one way or another, contributed to this achievement. Your encouragement, collaboration, and shared knowledge have made this journey both enriching and fulfilling.

Thank you all for being part of this remarkable experience.

ABSTRACT

The " E-Commerce Platform for Local Artisans to Showcase and Sell Handcrafted Products " project aims to empower artisans and craftsmen by providing them with a digital platform to showcase and sell their handmade products to a global audience. Artisans play a pivotal role in preserving cultural heritage and fostering economic sustainability in many communities. However, they often face challenges in reaching wider markets and selling their unique creations. The project offers a comprehensive e-commerce solution that bridges the gap between artisans and consumers. This platform provides artisans with the tools and resources needed to create online storefronts, manage their inventory, process orders, and receive payments securely. Customers can explore a diverse range of artisanal products, connect directly with the creators, and make informed purchases, thus supporting local craftsmanship and promoting sustainable consumption.

Keywords: Web application, E-commerce, Handicrafts, Ecommerce Local Artisans, online shopping website

TABLE OF CONTENTS

DECLARATION i

ACKNOWLEDGEMENT ii

ABSTRACT iii

LIST OF FIGURES vii

LIST OF TABLES viii

CHAPTER ONE 1

INTRODUCTION AND PROBLEM DESCRIPTION 1

1.1 Background 1

1.2 Problem Statement 1

1.3 Project Objectives 2

1.3.1 General Objective 2

1.3.2 Specific Objectives 2

1.4 Significance and Scope 3

1.4.1 Significance 3

1.4.2 Scope 3

1.4.3 Marketing Strategy 4

1.4.4 Proposed Technologies 5

CHAPTER TWO 6

LITERATURE REVIEW 6

2.1 Topic Review 6

2.2 Domain Review 6

2.2.1 Current Trends and Practices in E-Commerce for Artisans 6

2.2.2 E-Commerce Platforms for Artisans 7

2.2.3 Challenges Faced by Local Artisans 8

2.3 Problem Conclusion 9
CHAPTER THREE 10
REQUIREMENT ELICITATION AND SYSTEM ANALYSIS 10
3.1 Introduction to Requirement Elicitation 10
3.1.1 Requirement Elicitation 10
3.1.2 Data Collection Techniques 10
3.2 System Functionalities 11
3.2.1 Functional Requirements 11
3.2.2 Non-Functional Requirements 14
3.2.3 User Requirements 15
3.3 System Analysis 16
3.3.1 Competitive Analysis 16
3.3.3 System Models 18
CHAPTER FOUR 20
SYSTEM DESIGN 20
4.1 Introduction to System Design 20
4.2 System design methodology and technologies 21
4.2.1 Agile Methodology 21
4.2.2 Advantages of methodology 21
4.2.3 Technology and Tools 22
4.3 Database design 22
4.4 User Interface Design 25
4.4.1 Design Principles 25
4.4.2 Wireframes / Mockups 25
4.4.3 Tools Used for UI Design 26

CHAPTER FIVE 27
SYSTEM IMPLEMENTATION 27
5.1 Introduction to System Implementation 27
5.2 Functionalities Implementation 27
5.3 Database Implementation 30
5.4 User Interface Implementation 31
5.5 System Testing and Evaluation 32
5.5.1 Unit Testing 33
5.5.2 System Testing 33
5.5.3 Functional Testing 34
5.5.4 System Evaluation 34
CHAPTER SIX 35
CONCLUSION AND RECOMMENDATIONS 35
6.1 Summary of Achieved Objectives 35
6.2 Conclusion 35
6.3 Recommendations 36
REFERENCES 37

LIST OF FIGURES

Fig. 1.0 Showing use case diagram……22

Figure 2.0 Agile development life cycle 21

Fig. 3.0 E-Commerce Platform for Local Artisans ER Diagram..... 24

Fig. 4.0 Registration Page 27

Fig. 5.0 Product Listing Form..... 28

Fig. 6.0 Shopping Cart Page 29

Fig. 7.0 Checkout Page..... 29

Fig. 8.0 Order History Page 30

Figure 9.0 Ecommerce Platform for Local Artisans database implementation..... 31

Fig. 10 Artisan Dashboard 32

Fig. 11 Customer Dashboard 32

Fig. 12. Login Page..... 33

LIST OF TABLES

Table 1.0: Recommended technologies for implementing and deploying the E-commerce platform for Local Artisans 5

Table 2.0: Existing versus proposed platforms analysis 17

Table3.0: Show technologies used for implementing and deploying the E-commerce platform for Local Artisans 22

Table 4.0 Database Entities and Relationships 23

Table 5.0: Show the design principles used.....25

Table 6.0: Show tools used for user interface design. 26

CHAPTER ONE

INTRODUCTION AND PROBLEM DESCRIPTION

1.1 Background

The rise of industrial mass production has significantly impacted traditional craft markets, making it increasingly difficult for artisans to sustain their businesses. With the availability of cheaper, factory-made products flooding the market, many artisans struggle to compete and attract customers who appreciate unique, handcrafted items. As a result, traditional methods of selling, such as local craft fairs and physical stores, are becoming less viable for artisans who need a wider customer base to maintain profitability.

While existing e-commerce platforms such as eBay, Jiji Tanzania, U-buy Tanzania, and Kikuu primarily cater to mass-produced goods such as electronics and fashion items, there is a noticeable gap of dedicated platforms in Tanzania that support directly handmade and locally crafted products. Artisans attempting to sell their products on these platforms struggle with product visibility**,** high competition from factory-made goods, and platform fees that reduce their earnings. This gap has created a demand for an online marketplace that specifically connects Tanzania's artisans with a broader audience, allowing them to showcase their skills and sell their products with ease. To address this need, I propose an E-Commerce Platform for Local Artisans to Showcase and Sell Handcrafted Products, which will serve as a bridge between artisans and potential buyers. Through this platform, customers can explore a diverse collection of handmade products, interact directly with artisans, and make secure transactions. Ultimately, the goal is to provide artisans with a sustainable digital presence, empowering them to thrive in an increasingly digital economy.

1.2 Problem Statement

Many local artisans in Tanzania struggle to expand their businesses due to reliance on traditional selling methods like physical markets and word-of-mouth referrals. Despite the increasing global demand for handcrafted products, these artisans face limited market reach, making it difficult to sustain their livelihoods.

Although mainstream e-commerce platforms like Kikuu, eBay, and Jiji Tanzania provide opportunities for online selling, they primarily cater to mass-produced commercial goods rather

than handcrafted products. Artisans who attempt to use these platforms often face low visibility, high listing fees, and challenges in differentiating their products from factory-made alternatives.

Without an accessible, dedicated marketplace for handcrafted goods, local artisans are economically disadvantaged, and cultural craftsmanship is at risk of fading. This project aims to solve these challenges by developing an E-Commerce Platform for Local Artisans—a specialized digital marketplace where artisans can showcase and sell their handcrafted products to a broader audience, ensuring sustainable growth and financial empowerment.

1.3 Project Objectives

1.3.1 General Objective

The primary objective of this project is to develop an e-commerce platform that connects local artisans with a wider customer base, enabling them to showcase and sell their handcrafted products online. The platform aims to provide a seamless and user-friendly experience for both artisans and customers, fostering growth and sustainability in the artisanal industry.

1.3.2 Specific Objectives

The specific objectives of the project include:

To develop a digital marketplace for artisans – Creating an intuitive platform where artisans can register, create profiles, and list their handcrafted products with detailed descriptions and images.

To enhance market accessibility for artisans – Providing artisans with an opportunity to reach a larger audience beyond their local markets, expanding their customer base and sales potential.

To facilitate secure and convenient transactions – Implementing a reliable payment system that ensures safe and smooth financial transactions for both artisans and buyers.

To provide a personalized shopping experience for customers – Incorporating features such as search filters, product recommendations, and customer reviews to enhance user engagement and satisfaction.

To support artisan business growth – Equip artisans with tools to manage their inventory and track sales.

To build a community-driven platform where artisans and buyers can interact, share feedback, and appreciate the uniqueness of handmade products.

1.4 Significance and Scope

1.4.1 Significance

The proposed e-commerce platform holds significant value for both artisans and consumers. By providing local artisans with a dedicated digital marketplace, the platform empowers them to expand their reach beyond physical markets and gain financial stability. It addresses the growing demand for unique, handcrafted products by offering customers a centralized space to discover and purchase high-quality artisanal goods. Furthermore, the platform contributes to the preservation of traditional craftsmanship by ensuring that artisans can continue their trade in a competitive digital landscape.

From an economic perspective, the project fosters entrepreneurship by creating new income opportunities for artisans, thereby supporting local economies. It also enhances consumer awareness and appreciation for handmade products, encouraging sustainable consumption practices. By integrating features such as secure transactions, and customer feedback, the platform not only facilitates sales but also helps artisans refine their marketing and production strategies.

1.4.2 Scope

The platform will cater primarily to local artisans who specialize in handmade products such as textiles, jewelry, pottery, home décor, and other craft-based goods. It will allow artisans to register, create profiles, upload product listings, and manage sales transactions. Customers, on the other hand, will be able to browse products, interact with artisans, make secure purchases, and leave reviews.

Key features of the platform will include:

User Registration & Profiles: Separated profiles for artisans and customers with customizable dashboards.

Product Listings & Categories: Organized displays of handcrafted goods with detailed descriptions and pricing.

Secure Payment Integration: Support for various payment methods to ensure safe transactions.

Order Management System: Tools for artisans to track sales, manage inventory, and update product availability.

Customer Engagement Features: Messaging, reviews, and recommendations to enhance user interaction.

Mobile Responsiveness: Ensuring accessibility across different devices for a seamless shopping experience.

The platform will initially target artisans and consumers in Tanzania but has the potential for expansion into broader markets. Future enhancements may include:

Addition of new product categories such as home furnishings**.**

Implementation of a subscription model for users to receive price-drop notifications and newsletters.

Implementing business analytics that helps artisans refine their marketing and production strategies.

1.4.3 Marketing Strategy

To encourage artisans to join the platform, the following strategies will be implemented:

Free Registration for Early Users**:** Artisans who join within the first few months will have free access to premium features.

Partnerships with Local Handicraft Associations: Collaborations with artisan groups such as TanCraft will help onboard skilled sellers.

Social Media Advertising: The platform will leverage Facebook, Instagram, and TikTok ads to attract both artisans and buyers.

Influencer Marketing: Working with local influencers to promote artisan products and drive engagement.

Workshops and Training: Offering free training sessions to help artisans maximize their sales potential on the platform.

1.4.4 Proposed Technologies

Due to the nature of the project, desired technologies are those that:

Have been in active development, and have a wide and active user community,

Have passed the test of time,

Support and enforce good software engineering practices,

Are standards compliant,

Are easy to learn and use, and

Offer all the features required in the project.

Based on these qualities, the technologies shown in Table 1 are recommended.

Table 1.0: Recommended technologies for implementing and deploying the E-commerce platform for Local Artisans

Category	Technologies
Frontend Development	**HTML, CSS (TailwindCSS), JavaScript (React.js)**
Backend Development	**Python (Django), RESTful APIs, WebSockets** (e.g., Socket.IO)
Database Management	PostgreSQL

CHAPTER TWO

LITERATURE REVIEW

2.1 Topic Review

E-commerce platforms specifically designed for artisans play a crucial role in connecting local creators with global markets, providing them with a digital space to showcase handmade products, manage sales, and reach potential customers effectively (Miller et al., 2020). These platforms not only support artisans in marketing their crafts but also offer integrated tools for order management and secure payment processing, thereby empowering artisans to thrive in a competitive digital marketplace (Das & Bhattacharya, 2019). In many ways, these e-commerce platforms bridge the gap between traditional craftsmanship and the modern digital economy by enabling artisans to maintain the authenticity of their work while expanding their reach (Das & Bhattacharya, 2019). This review highlights the importance of these platforms and discusses how they preserve traditional crafts while promoting sustainable livelihoods for artisans.

2.2 Domain Review

This project falls within the domain of e-commerce, specifically focusing on online marketplaces for handcrafted products. E-commerce, in general, has revolutionized the retail industry by providing businesses and individuals with the ability to reach a global audience, breaking down barriers related to location, scale, and market access (Laudon & Traver, 2021). However, most conventional e-commerce platforms have been designed to cater to mass-produced goods, which can marginalize products with unique artistic value. There has been a growing demand for specialized online spaces that focus on handcrafted products, recognizing the value of artistry, uniqueness, and cultural significance embedded in these goods (Kaplan & Haenlein, 2020).

This section will explore current trends in the e-commerce marketplace for artisans, including the challenges they face and the importance of dedicated platforms to support their growth and sustainability. The proposed project focuses on building a platform specifically for artisans, with features designed to help overcome the limitations they face on generic e-commerce platforms.

2.2.1 Current Trends and Practices in E-Commerce for Artisans

Over the past decade, online marketplaces for handmade goods have surged, creating new opportunities for artisans to reach wider audiences while preserving the authenticity of their crafts.

Platforms such as Etsy, Amazon Handmade, eBay, and ArtFire have become central to this trend, providing a virtual space for artisans to showcase and sell their work (Huang & Benyoucef, 2013). These platforms provide not only retail outlets but also offer tools to tell the stories behind products, enhancing the emotional connection between the maker and consumer. This shift is particularly important as the market for personalized, unique, and artistic products grows in popularity, setting artisan-focused e-commerce platforms apart from traditional retail websites, where price and quantity are prioritized over uniqueness.

The rise of social commerce, leveraging platforms such as Instagram, Facebook, and Pinterest, has further amplified the reach of handcrafted goods. Artisans increasingly use these platforms to market their products and engage directly with their audience, creating a community around their products. This enables artisans to not only advertise their goods but also interact with potential customers, building relationships that go beyond the transactional (Olanrewaju et al., 2020). Social commerce allows artisans to blend marketing with sales, turning platforms that were once used primarily for content sharing into effective sales channels.

In Tanzania and other developing regions, this trend is particularly significant for small businesses and independent artisans who lack the resources for traditional advertising. By integrating social media with e-commerce, artisans can build an audience without investing in expensive marketing campaigns, making this approach highly effective for growing their digital presence.

2.2.2 E-Commerce Platforms for Artisans

Research into specialized e-commerce platforms for artisans reveals the critical role digital solutions play in addressing limitations faced by artisans on generic platforms. Many artisans report frustrations with the lack of customization on mainstream platforms, which makes it difficult for their unique products to stand out. Dedicated platforms, on the other hand, provide features specifically tailored to artisans' needs, such as customizable store profiles, integrated storytelling elements, and enhanced customer engagement tools. This allows artisans to build their brand identity and engage deeply with their customers.

"E-Commerce Web Application for Local Artisans": This study identifies the challenges artisans face on general-purpose e-commerce platforms, such as inadequate exposure and the lack of personalization. The study proposes a model for an e-commerce web application that provides artisans with a dedicated space for showcasing their products. It emphasizes the importance of a

tailored user experience that allows artisans to build their brand identity and engage with customers on a deeper level.

"EPICRAFT: E-commerce for Artisans": In this research, the focus is on developing an e-commerce platform using the MERN stack, a popular framework for building scalable web applications. The project aims to support India's handicraft and handloom industry by providing an accessible online marketplace for artisans. The study highlights the potential of leveraging modern technologies to create an efficient, scalable solution that empowers artisans to sell their work globally.

"Ecommerce Platform to Advertise Ghanaian": This study focuses on the creation of an e-commerce platform to promote Ghanaian artisans' products, integrating cultural storytelling with e-commerce functionalities. The research emphasizes the importance of localization and cultural representation, allowing Ghanaian artisans to reach a broader audience while educating consumers about the cultural significance of their handmade crafts. By providing artisans with a platform for direct sales, this approach fosters local economic development and encourages cultural exchange. The study highlights how digital platforms can empower artisans, enabling them to control their product narrative and improve their economic standing.

2.2.3 Challenges Faced by Local Artisans

Despite the opportunities provided by online platforms, artisans still face significant barriers that hinder their success. These challenges range from technological issues to cultural barriers and the struggle to differentiate their products from mass-produced goods. Understanding these challenges is key to developing solutions that will empower artisans to thrive in the digital marketplace.

Several studies highlight the challenges artisans face in leveraging existing e-commerce platforms:

Limited Digital Literacy: Many artisans lack the necessary skills to effectively manage an online store, including uploading product images, writing descriptions, and handling customer inquiries (Hussain et al., 2020).

High Transaction Fees: Popular platforms impose listing and transaction fees that reduce artisans' profit margins (Kumar & Verma, 2019).

Marketing and Visibility Issues: Standing out in highly competitive marketplaces is difficult, particularly for artisans with limited financial resources for digital marketing (Rodriguez, 2021).

Trust and Payment Security: Many artisans hesitate to engage in online transactions due to concerns over fraudulent buyers and lack of secure payment methods (Nguyen, 2022).

Logistics and Delivery Challenges: Reliable logistics solutions are essential for successful e-commerce, yet many Tanzanian artisans struggle with high shipping costs and inefficient delivery systems (Moshi, 2023).

2.3 Problem Conclusion

The literature suggests that e-commerce platforms have the potential to empower artisans by providing digital tools to showcase and sell their products globally. However, existing solutions often present barriers such as high costs, lack of customization, and visibility challenges. The proposed platform seeks to bridge these gaps by offering an accessible, cost-effective, and artisan-friendly e-commerce solution, ensuring ease of use while enhancing market reach. By leveraging Agile methodology, the system will be designed iteratively, incorporating feedback from Tanzanian artisans to create a sustainable and impactful solution.

CHAPTER THREE

REQUIREMENT ELICITATION AND SYSTEM ANALYSIS

3.1 Introduction to Requirement Elicitation

The requirement is a statement that identifies the capability, characteristics, or quality of the system for it to have value and utility to a customer or user (Young, 2003). Elicitation is the gathering of all system requirements from stakeholders and encompasses all activities involved in discovering the requirements of a system.

Requirement elicitation and analysis is the process of interacting with customers and end-users to find out about the domain requirements, what services the system should provide, and other constraints (Sommerville, 2011). This phase is crucial in the software development lifecycle as it involves gathering, analyzing, and defining stakeholders' needs to ensure the successful development of the e-commerce platform.

To develop a user-friendly and efficient platform, various elicitation techniques will be employed, including interviews with artisans and potential buyers, surveys, observations, and analysis of existing e-commerce platforms. Understanding the challenges artisans face in accessing digital markets and the expectations of customers helps shape the platform's features, such as product listing, secure payments, inventory management, and customer interaction. Proper requirement elicitation ensures that the final system aligns with user needs, enhances business growth for artisans, and delivers a seamless shopping experience for customers.

3.1.1 Requirement Elicitation

3.1.2 Data Collection Techniques

These are the techniques that describe how tasks are performed or how data are collected under specific circumstances. Collecting requirement are based in various ways which aim to understand the system's needs, how it will operate, who are the customer or users of the system, and what product fits the needs of the customers.

The following techniques will be employed to gather system requirements:

Interviews – The method involve discussions with local artisans, craft business owners, and potential customers using both face-to-face (personal) interview and telephone interview to understand their challenges and expectations from the platform.

Questionnaires – The method involves distributing Surveys to artisans and customers to collect insights on their preferences and concerns regarding an online marketplace.

Observation – The method involves analyzing how artisans currently sell their products, including challenges faced in physical markets and social media sales.

Document Review – The method involves examining existing online marketplaces and business reports on e-commerce adoption among artisans.

3.2 System Functionalities

3.2.1 Functional Requirements

The functional requirements are specific, measurable, and testable capabilities that a system or product must possess to meet the needs of its stakeholders.

These requirements typically detail the functions that a system must perform, such as input and output, data processing, and user interactions.

The e-commerce platform for artisans incorporates several key functionalities to enhance usability, efficiency, and security. These functionalities include:

1. User Management:

Registration and Login System**:**

Artisans and customers must be able to register and log in using their email, social media accounts (optional), or phone number.

Registration should be validated via email or phone for account verification.

Multi-factor authentication (MFA) will be implemented for additional security for users.

Profile Management for Artisans**:**

Artisans should be able to create, edit, and manage their personal profiles.

Each artisan profile should include a bio, shop name, contact information, and an image gallery to showcase their work.

Artisans can manage and update their password and other personal settings.

The artisan's profile page will display their product listings and customer reviews.

2. Product Management:

Add, Edit, and Delete Products**:**

Artisans must be able to add new products by uploading images, adding detailed descriptions, setting prices, and categorizing them based on product type or theme.

Artisans can edit and update product details as needed, including changing prices, stock levels, or images.

Artisans can delete products that are no longer available for sale.

Categorization of Products**:**

Products should be categorized into distinct types (e.g., jewelry, pottery, clothing) to enhance user experience and make searching easier.

A tagging system could be used to allow products to be filtered by attributes such as color, material, size, or style.

3. Shopping and Checkout:

Product Browsing and Search**:**

Customers can browse the available products based on categories, tags, or specific search terms.

A robust search feature must allow filtering of products based on price range, type, or rating.

A product recommendation system can be implemented to suggest products based on browsing history.

Shopping Cart**:**

Customers can add multiple products to their shopping cart before proceeding to checkout.

Users should be able to edit the cart by adding or removing items, adjusting quantities, and reviewing the total cost.

Checkout Process**:**

Customers should be able to proceed with their orders, enter shipping details, and select payment methods (e.g., card payment, mobile money, or bank transfer).

Secure integration with a payment gateway (such as Pesapal) should be implemented to process payments.

The platform must securely store order information, customer details, and payment history.

4. Order and Sales Tracking:

Order History for Customers**:**

Customers must have access to their order history, which includes order dates, item details, status, and payment information.

Customers should be able to track the status of their orders, such as "Processing," "Shipped," or "Delivered."

Notifications for Artisans**:**

Artisans should receive real-time notifications when new orders are placed, allowing them to manage their inventory and fulfill orders promptly.

The system should allow artisans to mark orders as completed once they are shipped or delivered.

Sales Reports for Artisans**:**

Artisans will have access to detailed sales reports that include metrics such as total sales, best-selling products, and customer engagement.

Reports will be exportable in formats such as CSV or PDF for easy record-keeping.

5. Review and Rating System:

Customer Reviews for Products**:**

After purchasing an item, customers can leave reviews on the product, including a rating (1 to 5 stars) and a written comment.

Reviews will help future customers make informed decisions by displaying an average rating and recent reviews for each product.

Rating Visibility**:**

Products with higher ratings will be prioritized in search results, creating an incentive for artisans to maintain high product quality.

A mechanism will allow customers to report inappropriate reviews or ratings.

6. Messaging and Notifications:

Artisan and Customer Communication**:**

Artisans and customers can communicate through a secure messaging system within the platform.

The messaging system should allow both parties to discuss orders, clarify product details, and resolve any issues directly.

Automated Notifications**:**

Automated notifications should be sent to customers for key events such as order confirmation, shipment tracking, and delivery status.

Artisans should be notified when their products are purchased, when an order is shipped, and when a review is posted.

Promotional notifications (e.g., discounts, new products, or sales) should be customizable by the platform administrators and available to both artisans and customers.

3.2.2 Non-Functional Requirements

These are constraints on the services or functions offered by the system. They include timing constraints, constraints on the development process, and constraints imposed by standards. Non-functional requirements often apply to the system as a whole, rather than individual system features or services (Sommerville 2011).

The system must adhere to the following quality attributes:

Security: The system should ensure Data encryption, user authentication, and secure payment transactions.

Performance: The system should be fast and responsive, with minimal delays for end-users.

Usability: The system should be user-friendly, with a clear and intuitive interface that is easy to navigate for artisans and customers.

Scalability: The system should have ability to support an increasing number of users and products.

Compliance: The system should comply with relevant regulations and standards, such as data privacy and security regulations.

Interoperability: The system should be capable of integrating with other systems, such as payment gateways, accounting software, and reservation systems.

Maintainability: The system should be easy to maintain and upgrade, with clear documentation and a modular design.

Portability: The system should be capable of running on different platforms and hardware configurations.

3.2.3 User Requirements

The E-Commerce Platform for Local Artisans should meet the functional needs of different users, including customers (buyers), artisans (sellers), and administrators.

The user requirements are as follows:

The system should allow users (customers and artisans) to register and log in securely.

Customers should be able to browse products listed by artisans, search for items based on categories, price, or artisan name, and view detailed product information, including descriptions, images, and pricing.

The platform should enable customers to add products to a shopping cart, modify their cart before checkout, and complete purchases using a secure payment gateway.

The platform should provide order tracking, allowing customers to check the status of their purchases, while artisans should be able to receive notifications for new orders, view order details, and update the order status (e.g., processing, shipped, delivered).

Artisans should have the ability to create and manage product listings by adding images, descriptions, and pricing, as well as updating stock levels or removing products when necessary.

The system should facilitate direct communication between customers and artisans, allowing artisans to respond to customer inquiries regarding their products.

Customers should have the ability to leave reviews and ratings for purchased products, while artisans should be able to view feedback to enhance their offerings.

The platform should generate sales performance and earnings reports for artisans, enabling them to track their business growth.

Administrators should have the ability to approve or reject artisan account registrations, manage user accounts (both buyers and sellers), and oversee product listings to ensure compliance with platform policies.

Admins should have access to customer complaints and be able to resolve disputes between buyers and sellers to maintain trust and fairness on the platform.

3.3 System Analysis

System Analysis refers to a problem-solving technique that decomposes a system into its component pieces for the purpose of studying how well those component parts work and interact to accomplish their purpose. This is a process used in the design of new systems. Systems analysis follows stages of investigation, design and implementation.

3.3.1 Competitive Analysis

The proposed E-Commerce Platform for Local Artisans differs from existing solutions such as Etsy, Jiji Tanzania, Jumia Tanzania, Kikuu, Amazon Handmade, and Spring Soko in the following ways:

Table 2.0: Existing versus proposed platforms analysis

Platform	**Key Features**	**Limitations**	**How the Proposed System Addresses These Gaps**
Etsy	Global marketplace for handmade and vintage goods.	High competition for local artisans, expensive listing fees, and international shipping constraints.	Focuses on Tanzanian artisans only, with lower fees and localized delivery methods.
Jiji Tanzania	Online classified platform for selling various products.	No direct e-commerce transactions, no buyer protection, lacks structured artisan marketplace.	Implements a direct checkout system with integrated payment and buyer protection.
Kikuu	Imports affordable goods from China to African markets.	No support for local artisans, mostly resellers of mass-produced goods.	Prioritizes locally made crafts, promoting Tanzanian culture and craftsmanship.
eBay	Global marketplace for various products, including handmade goods.	High seller fees, limited support for artisan branding, and intense competition.	Provides dedicated branding tools for artisans, lower listing fees, and localized payment methods.

The proposed system offers lower costs, a local focus, mobile money payments, and a strong artisan community, making it more accessible and sustainable for Tanzanian craftsmen.

3.3.3 System Models

To illustrate the system's structure, the following model will be used:

Use Case Diagram

A use case diagram is a type of Unified Modelling Language (UML) diagram that depicts the interactions between a system and its actors to achieve a specific goal.

The main components of a use case diagram include:

Actors: These are external entities (users or other systems) that interact with the system. They are typically represented by stick figures or appropriate symbols.

System: This represents the application or process being modeled. It is depicted as a rectangle containing the system's name.

Use Cases: These represent specific functionalities or actions that the system provides to the actors. They are shown as ovals inside the system boundary.

Below is a Use Case Diagram for an E-Commerce Platform for Local Artisans to Showcase and Sell Handcrafted Products:

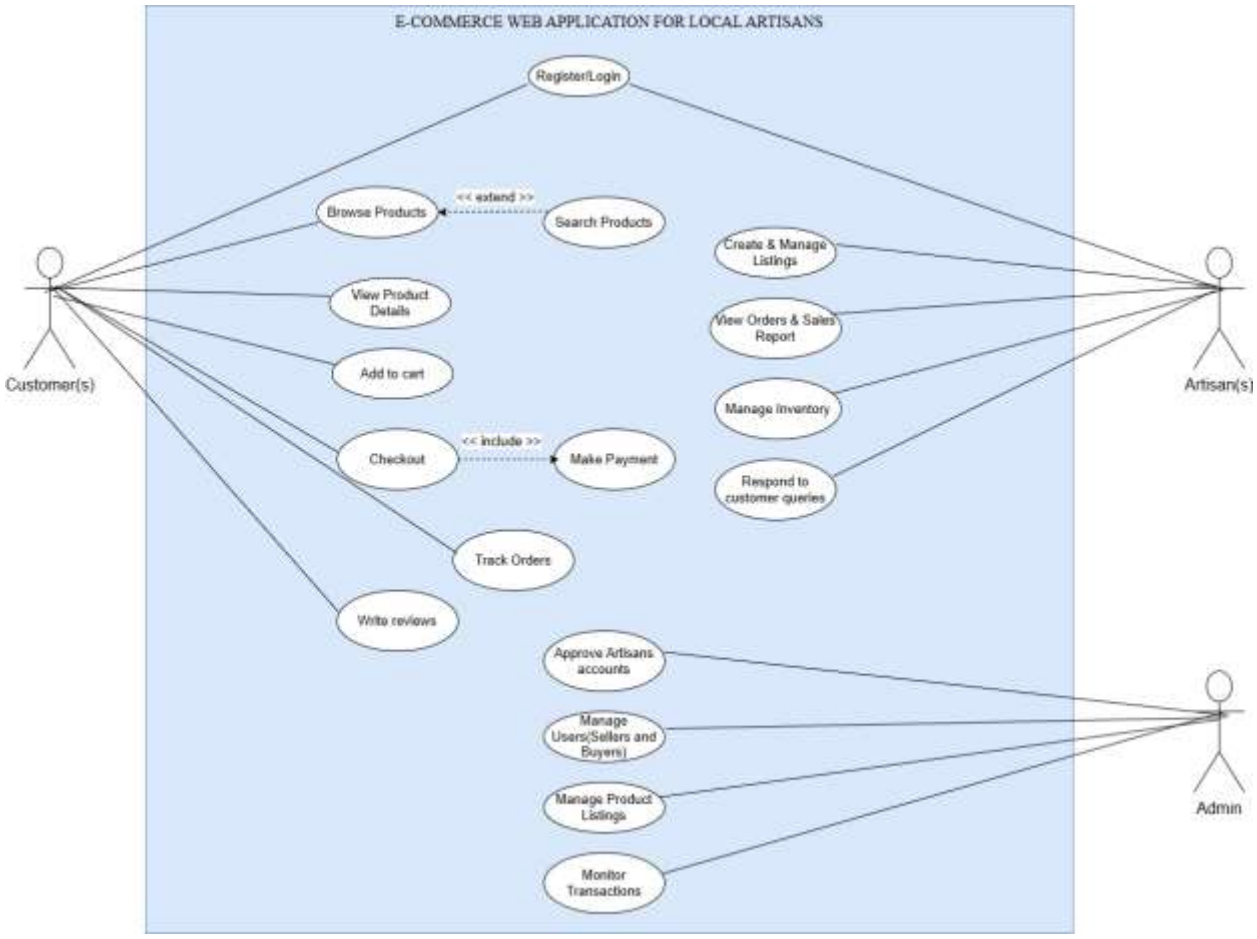

Fig. 1.0 Showing use case diagram

CHAPTER FOUR

SYSTEM DESIGN

4.1 Introduction to System Design

System design is a critical phase in the Software Development Life Cycle (SDLC) that translates user requirements and system functionalities into a structured and workable system architecture. It serves as the blueprint for the entire application, defining how the system's components will interact, how data will flow, and how the user will engage with the system.

In this chapter, the system architecture, design methodology, database schema, and user interface designs are presented. The goal is to ensure the system is scalable, secure, user-friendly, and aligned with the needs of local artisans and customers.

Types of System Design

System design can be categorized into three main levels, each focusing on a different aspect of system development:

Architectural Design: This level defines the high-level structure of the system, identifying major components, their responsibilities, and the relationships between them. It provides a conceptual framework that guides the implementation process and ensures coherence in system development.

Logical Design: Logical design is a technology-independent representation of the system's functionality and behavior. It emphasizes the processes, data flows, and business rules that govern system operations. At this level, the goal is to model what the system should do without getting into technical implementation details, making it easier to validate requirements and ensure stakeholder alignment.

Physical Design: Physical design translates the logical design into a concrete implementation. It focuses on the specific technologies, platforms, databases, and infrastructure required to deploy the system. This includes defining the software and hardware environments, network configuration, and data storage structures necessary to meet performance, security, and scalability requirements.

4.2 System design methodology and technologies

The system was developed using the Agile methodology, which emphasizes iterative development, collaboration, and responsiveness to user feedback. Agile was chosen because it supports continuous improvement and allowed regular input from potential users (artisans and customers) during development.

4.2.1 Agile Methodology

The Agile methodology is a way to manage a project by breaking it up into several phases. It involves constant collaboration with stakeholders and continuous improvement at every stage. Once the work begins, teams' cycle through a process of planning, executing, and evaluating. Continuous collaboration is vital, both with team members and project stakeholders.

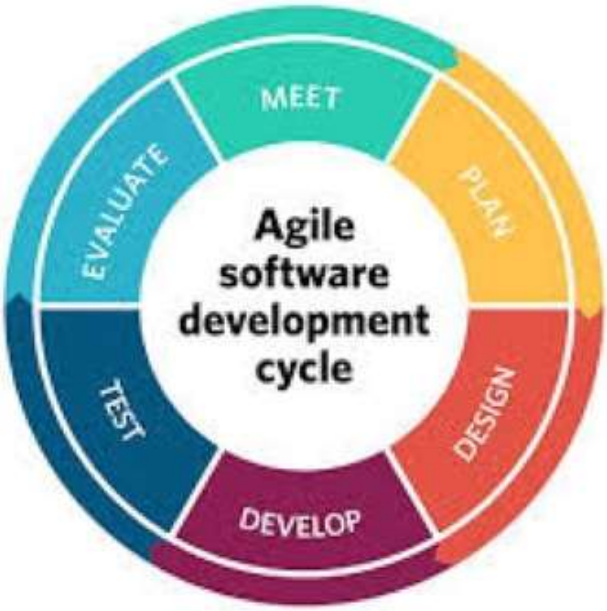

Figure 2.0 Agile development life cycle

4.2.2 Advantages of methodology

Agile methodology offers several key benefits:

Flexibility and Adaptability: It supports changing requirements and allows teams to adjust quickly throughout development.

Customer Satisfaction: Involving stakeholders regularly ensures the final product aligns with their needs and expectations.

Early and Continuous Delivery: Frequent delivery of functional product increments provides early value and feedback opportunities.

Improved Quality: Ongoing testing and feedback help detect and resolve issues early, leading to a more reliable system.

Risk Mitigation: Its iterative nature allows early identification and resolution of risks, minimizing potential project failure.

4.2.3 Technology and Tools

Table3.0: Show technologies used for implementing and deploying the E-commerce platform for Local Artisans

Component	**Technologies**	**Justification**
Frontend	React.js, Tailwind CSS	Dynamic UI with responsive design.
Backend	Django (Python)	Rapid development, built-in security (e.g., CSRF protection).
Database	PostgreSQL	ACID compliance for reliable transactions.
Authentication	JWT	Secure, stateless user sessions.

4.3 Database design

The database design is based on relational database principles to ensure data consistency, normalization, and integrity. The PostgreSQL database was used for its robustness and performance in handling transactional operations.

Key Tables and Relationships:

To outline a conceptual database model for our ecommerce platform for local artisans, the following entities was found:

Table 4.0 Database Entities and Relationships

Entity	Description
Customer	Users who register and place orders.
Product	Items available for sale.
Category	Groups products into logical classifications.
Order	Purchase records created by customers.
Order_Item	Specific items within an order.
Payment	Transaction details linked to orders.
Shipment	Delivery information for orders.
Cart	Temporary storage of selected products before purchase.
Wishlist	Customer-selected products for future consideration.

The relationships between the entities are as follows:

Customer → Order: One-to-many – a customer can place multiple orders.

Order → Order_Item → Product: Orders contain multiple items; each item links to a product.

Order → Payment / Shipment: Each order has one payment and one shipment; each payment/shipment may serve multiple orders.

Product → Category: One-to-many – each product belongs to one category.

Cart/Wishlist → Customer & Product: Both are linked to one customer and one product, forming many-to-one relationships with Product and dependency on Customer.

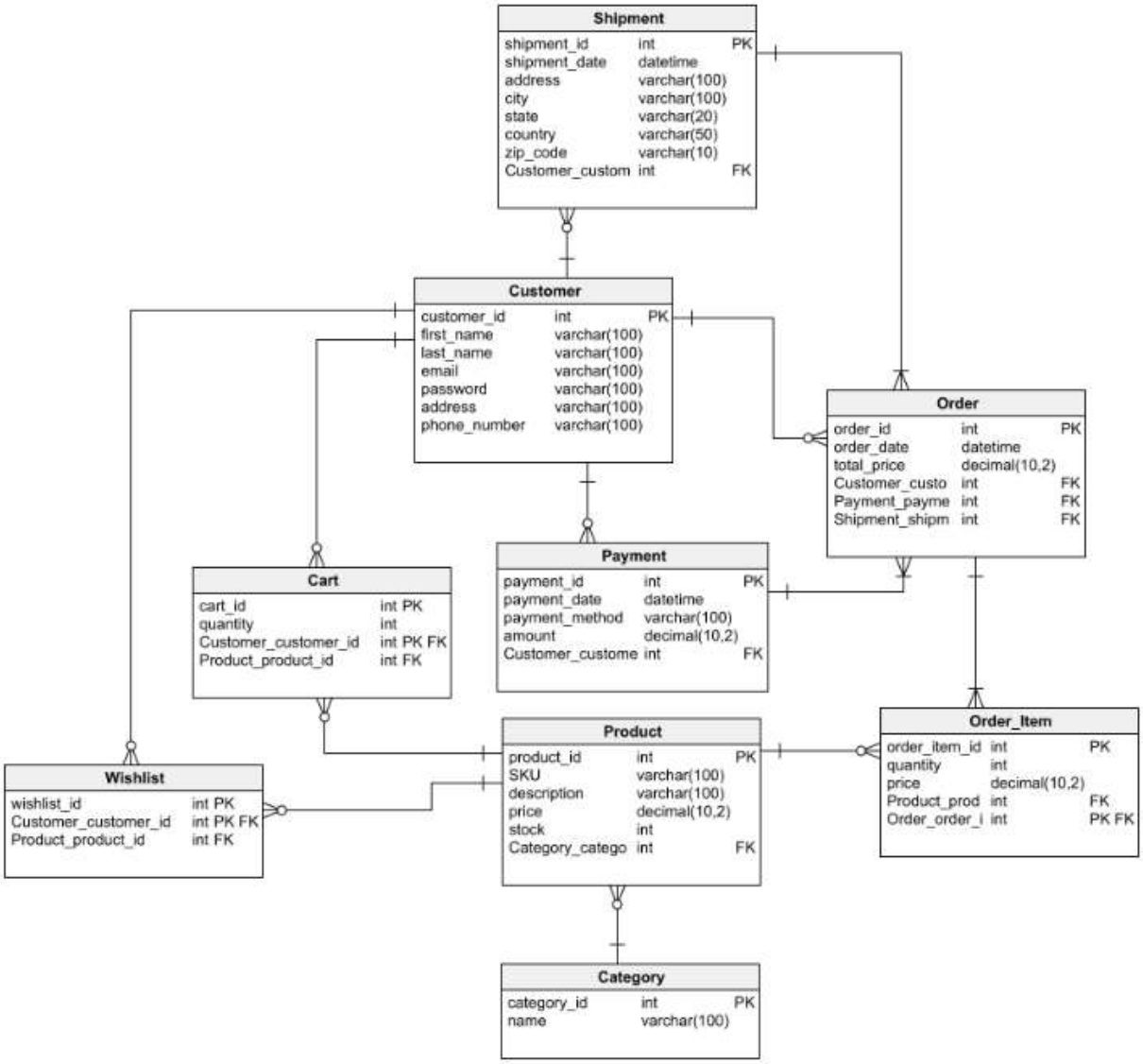

Fig. 3.0 E-Commerce Platform for Local Artisans ER Diagram

4.4 User Interface Design

The user interface (UI) is the bridge between the system and its users. It was designed to be clean, responsive, and user-friendly, focusing on accessibility for artisans with varying digital literacy levels.

4.4.1 Design Principles

The UI/UX design follows these core principles to ensure usability and engagement:

Table 5.0: Show the design principles used.

Principle	**Description**
User-Friendly Navigation	Simple menus, intuitive icons, and a search bar for quick product discovery.
Mobile Responsiveness	Responsive design ensures seamless use on smartphones, tablets, and desktops.
Visual Appeal	High-quality product images, consistent color schemes, and clean typography.
Accessibility	WCAG-compliant contrast ratios, alt text for images, and keyboard navigation.
Consistency	Uniform buttons, fonts, and layouts across all screens for familiarity.

4.4.2 Wireframes / Mockups

Key User Interface (UI) Components:

User Authentication Pages: Sign-up and login forms with validation for both artisans and customers.

Artisan Dashboard:

Features: Product upload form, sales tracking panel, order list, and profile management.

Customer Dashboard: Featured with Order history, saved products, personal profile settings.

Product Listing Page: Grid view of products with filters by category, price range, and ratings.

Product Details Page: Detailed description, artisan info, add-to-cart button, and customer reviews.

Checkout Page: Displays cart summary, allows input of delivery details, and integrates Pesapal for payment.

Messaging Interface: Allows customers and artisans to communicate directly regarding product inquiries or custom requests.

Admin Panel: User management, content moderation, and platform analytics dashboard

4.4.3 Tools Used for UI Design

Table 6.0: Show tools used for user interface design.

Tool	**Purpose**	**Justifications**
Figma	Prototyping wireframes and collaborative design iterations.	Enables real-time collaboration with stakeholders.
Tailwind CSS	Utility-first CSS framework for responsive styling.	Accelerates development with pre-built responsive components.

CHAPTER FIVE

SYSTEM IMPLEMENTATION

5.1 Introduction to System Implementation

System implementation is a crucial phase in the Software Development Life Cycle (SDLC) that focuses on transforming the designed system into a working solution. It involves translating designs and requirements into executable code, configuring databases, setting up the user interface, and ensuring all components work together as a cohesive application. This chapter presents how the functionalities, database, user interface, and testing processes were implemented for the E-commerce Platform for Local Artisans.

5.2 Functionalities Implementation

The system was implemented using a modular approach. Each key functionality outlined during system analysis was developed separately, then integrated to ensure cohesion and usability. The following modules were developed:

User Registration and Authentication: Artisans and customers can create accounts using email addresses. Django's built-in authentication system was extended with user roles to differentiate between artisans and buyers.

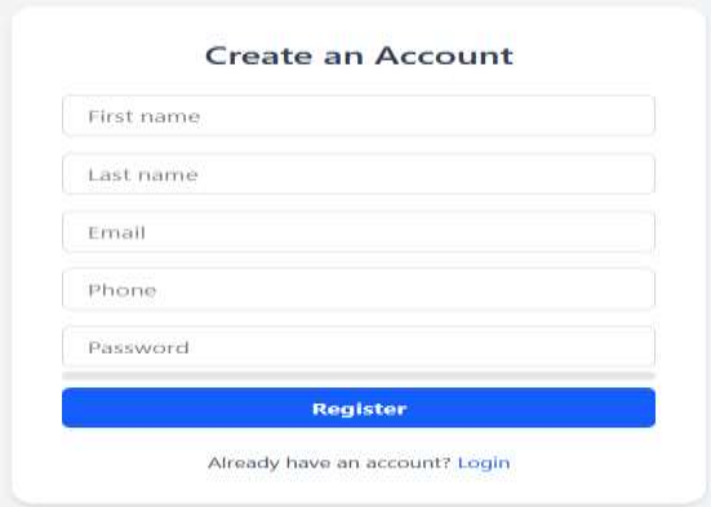

Fig. 4.0 Registration Page

Product Management Module: Allows artisans to upload product images, set prices, write descriptions, and categorize their products. CRUD (Create, Read, Update, Delete) functionalities were implemented using Django's class-based views.

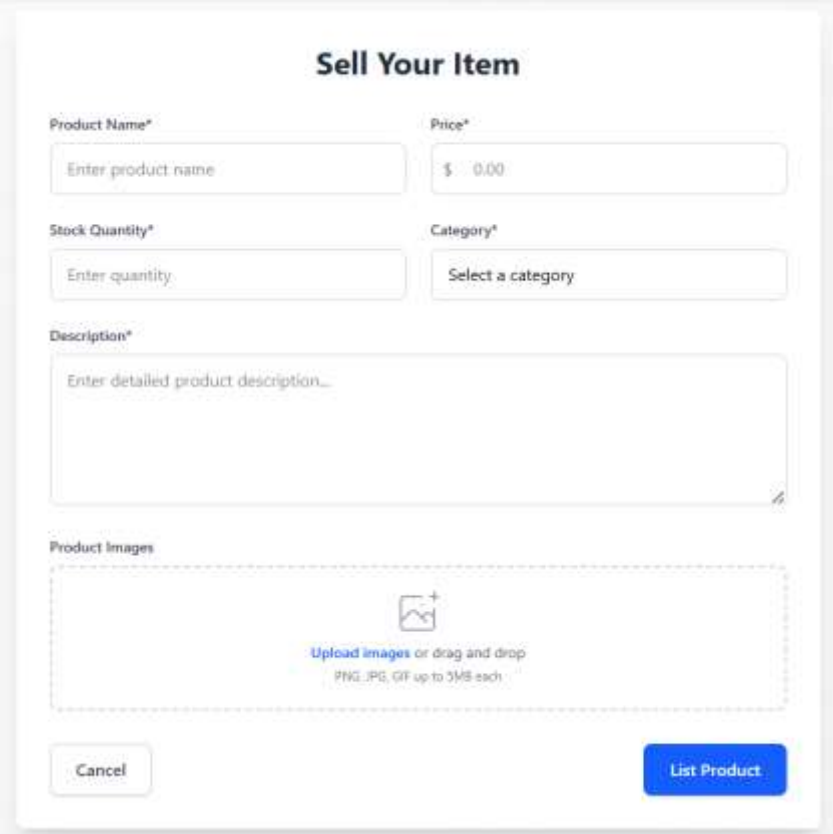

Fig. 5.0 Product Listing Form

Shopping Cart and Checkout: Customers can add items to a cart, view summaries, and proceed to checkout. Payment integration was implemented using **Pesapal,** enabling secure transactions via mobile money and cards.

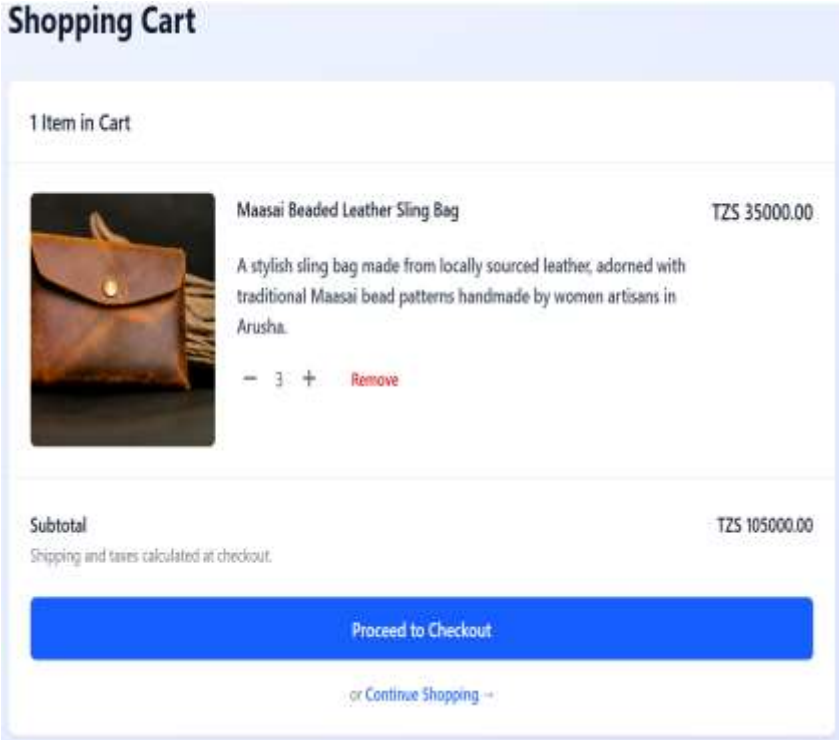

Fig. 6.0 Shopping Cart Page

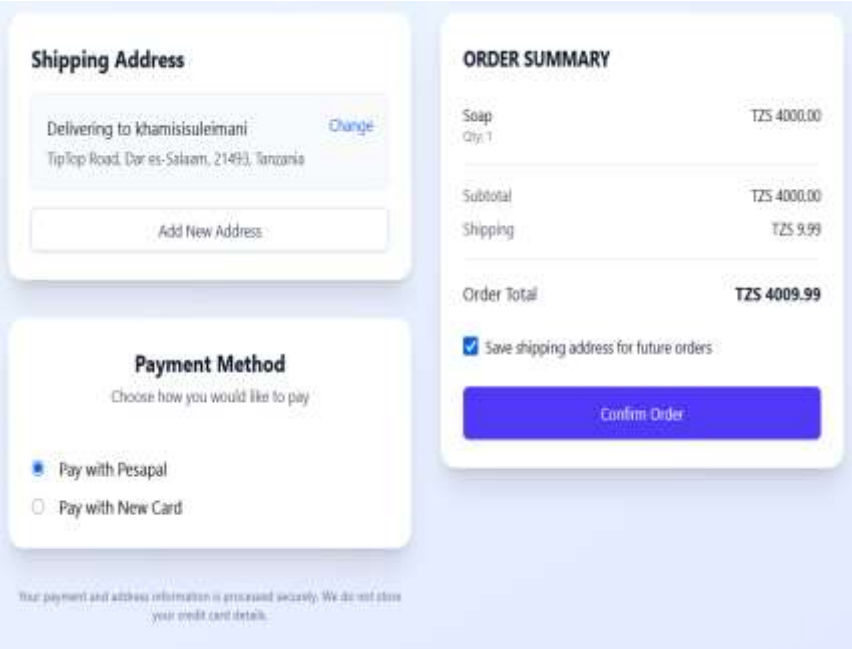

Fig. 7.0 Checkout Page

Order Management: Orders are tracked by both the artisan and the customer. Artisans can update the status (processing, shipped, delivered), while customers can view their order history.

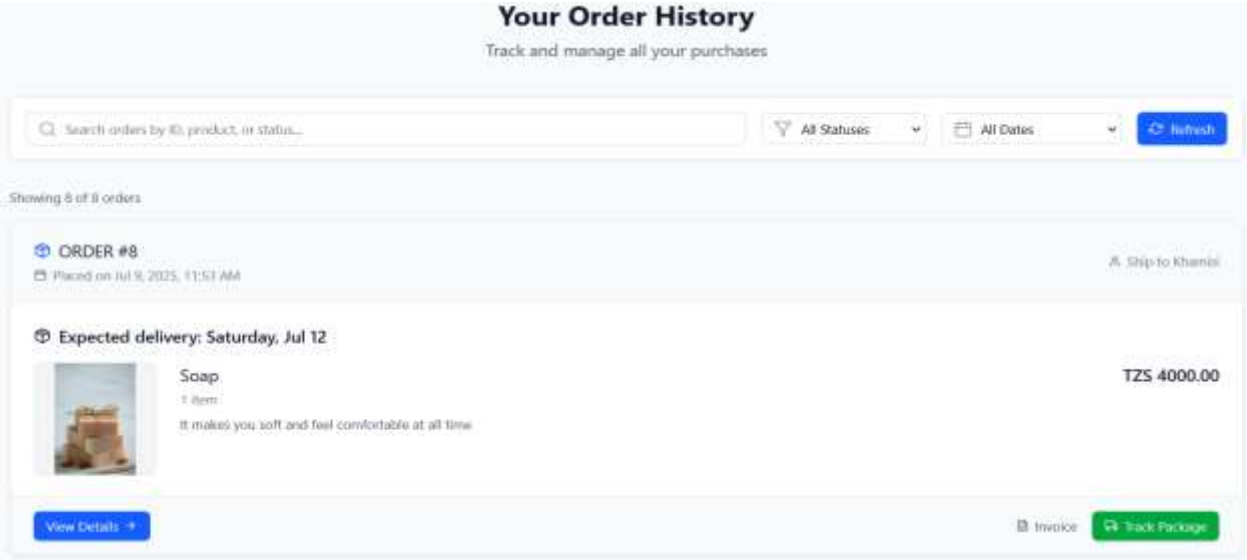

Fig. 8.0 Order History Page

Messaging and Notifications: A real-time messaging feature enables buyers and artisans to communicate. Notifications are sent on order confirmations, product delivery, and new messages.

Review and Rating System: After each purchase, customers can rate and review products, enhancing transparency and trust within the marketplace.

5.3 Database Implementation

The PostgreSQL database was chosen for its robustness and scalability. The database schema was normalized to eliminate redundancy and ensure data integrity.

In the Django project, the database configuration was defined in the DATABASES setting of the settings.py file:

ENGINE: 'django.db.backends.postgresql' - Specifies that Django will use the PostgreSQL backend.

NAME: 'artisan_app' - The name of the database created for this project.

USER and PASSWORD: Credentials used to connect to the database. (For security reasons, these are typically stored in environment variables, not hardcoded.)

HOST: 'localhost' - Indicates that the database runs on the same machine as the application.

PORT: '5432' - The default PostgreSQL port.

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql',
        'NAME': config('DB_NAME'),
        'USER': config('DB_USER'),
        'PASSWORD': config('DB_PASSWORD'),
        'HOST': config('DB_HOST'),
        'PORT': config('DB_PORT'),
    }
}
```

Figure 9.0 Ecommerce Platform for Local Artisans database implementation

Database migrations were managed using Django's Object-Relational Mapping (ORM) commands:

python manage.py makemigrations

python manage.py migrate

These commands ensured a smooth evolution of the database schema throughout the development lifecycle.

5.4 User Interface Implementation

The user interface (UI) was developed using React.js with Tailwind CSS, resulting in a responsive and modern look across devices. Key UI elements include:

Artisan Dashboard: Custom panel for artisans to manage products, view orders, and monitor earnings.

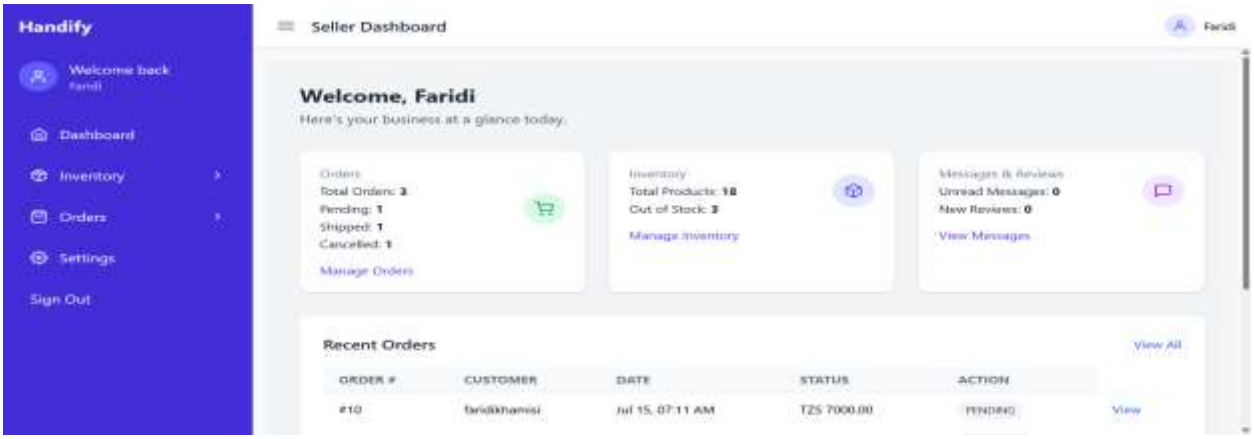

Fig. 10 Artisan Dashboard

Customer Dashboard: Provides access to order history, saved items, and messages. Displays detailed product information, image carousel, and related products.

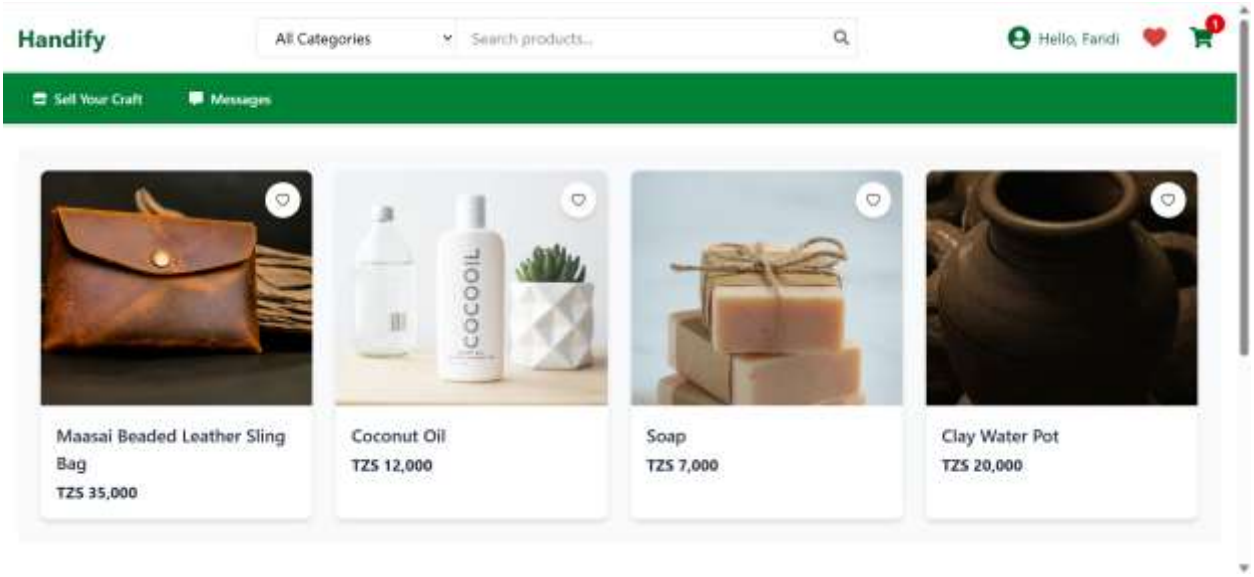

Fig. 11 Customer Dashboard

Design considerations focused on simplicity, mobile responsiveness, and visual appeal to ensure accessibility and user satisfaction.

5.5 System Testing and Evaluation

Testing was essential to ensure that all modules function correctly, securely, and efficiently. Multiple testing techniques were used throughout the development cycle.

5.5.1 Unit Testing

Unit testing focused on individual components and functions to verify correctness. Examples include:

Validation of login credentials and form fields.

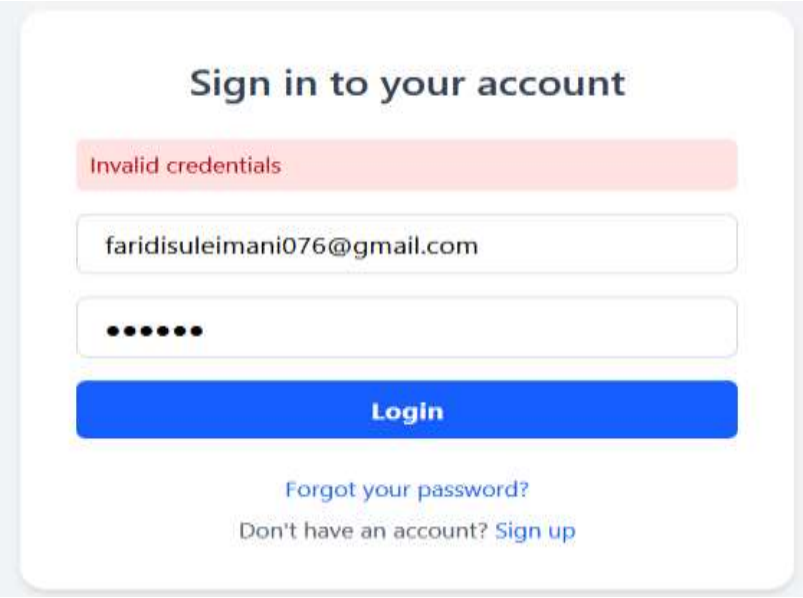

Fig. 12. Login Page

Testing product upload and update functionalities.

Payment processing logic using mock transactions.

Tools such as Django's built-in test framework (unittest) were used for automated unit testing.

5.5.2 System Testing

System testing evaluated the interaction of various modules to ensure the platform operates as a cohesive unit. Test cases included:

Registering new users and assigning correct roles.

Simulating a full order process from product selection to payment and delivery.

Ensuring data integrity across modules, such as product deletion affecting order records.

This testing confirmed that modules integrated correctly and business rules were upheld.

5.5.3 Functional Testing

Functional testing ensured that the system behaves according to the requirements. Each functional requirement was tested to verify:

Role-based access control (artisans, customers, admin).

Checkout and payment processing flow.

Notifications for order status updates.

All features passed functional testing and matched expected outcomes defined in the requirement specification.

5.5.4 System Evaluation

System evaluation focuses on assessing the performance, usability, security, and user satisfaction of the implemented platform. The evaluation process involved both technical assessment and user feedback from a selected group of artisans and customers.

To evaluate the system effectively, the following criteria and results were used:

Functionality: Most users confirmed that the system provided all essential features and operated reliably without experiencing significant errors or crashes.

Performance: The platform demonstrated consistently fast response times, with pages typically loading in under two seconds even when multiple users were active simultaneously.

Security: Testing revealed no security vulnerabilities. Key features such as password encryption and role-based access controls were implemented effectively and performed as expected.

User Satisfaction: A large majority of users reported a positive experience with the platform and expressed a willingness to recommend it to others.

CHAPTER SIX

CONCLUSION AND RECOMMENDATIONS

6.1 Summary of Achieved Objectives

The primary objective of this project was to develop an e-commerce platform tailored for Tanzanian artisans to digitally showcase and sell their handcrafted products. Through systematic planning, design, and implementation, the following objectives were successfully achieved:

Development of a digital marketplace for artisans: The system allows artisans to register, create detailed profiles, upload product listings with images, and manage their stores.

Enhanced market accessibility: Artisans can now reach a broader audience beyond their local communities, thereby increasing sales potential and visibility.

Secure and convenient transactions: Integration with Pesapal enables secure payments using mobile money and card options.

Personalized customer experience: Customers can search and filter products, read reviews, and interact directly with artisans.

Support for artisan business growth: Features such as order management, real-time notifications, sales tracking, and messaging help artisans manage and expand their businesses.

Community engagement: The platform supports ratings, reviews, and messaging, fostering a collaborative and transparent environment for buyers and sellers.

All outlined functional and non-functional requirements were implemented and verified through system testing and user evaluation, ensuring the platform meets its intended purpose.

6.2 Conclusion

The successful development and evaluation of the e-commerce platform signify a significant step toward empowering local artisans through digital technology. The platform bridges the gap between traditional craftsmanship and the modern digital marketplace, offering a sustainable solution for artisans to thrive economically.

By addressing core challenges such as limited market reach, lack of visibility, and inefficient selling channels, the platform enables artisans to showcase their unique creations to a wider

audience. Customers benefit by accessing authentic, handcrafted goods while supporting local economies and cultural preservation.

The system is scalable, user-friendly, and secure, with the potential to expand beyond Tanzania and accommodate more features as user demand grows. Ultimately, this project not only fulfills academic requirements but also contributes meaningfully to digital inclusion and economic empowerment for underserved communities.

6.3 Recommendations

Based on the findings and feedback during the system evaluation phase, the following recommendations are proposed for future improvements:

Localization Support: Incorporating Swahili and other regional languages to improve accessibility for artisans with limited English proficiency.

Mobile App Development: Developing a lightweight Android app to reach users with limited access to computers or stable internet connections.

Logistics and Delivery Integration: Establishing Partnership with local logistics providers to enable real-time delivery tracking and reduce shipping complexity.

Training and Digital Literacy Workshops: Organizing regular digital training sessions to help artisans better understand how to use the platform effectively.

Scalability to Other Markets: Plan for platform expansion across East Africa to support artisans in neighboring countries with similar needs.

By implementing these recommendations, the platform can evolve into a comprehensive solution that further empowers artisans, enhances customer experience, and fosters sustainable growth within the local and regional digital economy.

REFERENCES

Gursoy, D., Chi, O. H., & Kim, S. (2022). Artificial intelligence in e-commerce: The impact on consumer engagement and business performance. *Journal of Business Research*, 139, 812-823.

Huang, Z., & Benyoucef, M. (2013). User-centered design of e-commerce websites for the elderly. *Electronic Commerce Research and Applications*, 12(4), 261-272.

Kaplan, A., & Haenlein, M. (2020). Rethinking e-commerce: The role of social media in online shopping. *Business Horizons*, 63(1), 79-88.

Kim, J., Suh, J., & Lee, H. (2020). The impact of e-commerce on small businesses: A case study of artisan markets. *International Journal of Electronic Commerce*, 24(2), 245-268.

Laudon, K. C., & Traver, C. G. (2021). *E-Commerce 2021: Business, Technology, Society*. Pearson.

Niranjanamurthy, M., Kavyashree, N., & Kumar, D. (2019). Analysis of e-commerce ecosystem and its impact on artisan entrepreneurship. *Asian Journal of Business and Management*, 7(3), 112-125.

Nordin, R., Hashim, H., & Ismail, S. (2021). The role of secure payment systems in enhancing consumer trust in online artisan markets. *Journal of Retailing and Consumer Services*, 58, 102275.

Olanrewaju, A., Shah, P., & Bello, K. (2020). The influence of social commerce on the success of online handmade businesses. *Computers in Human Behavior*, 108, 106322.

Porter, M. E., & Kramer, M. R. (2019). Creating shared value: How to reinvent capitalism—and unleash a wave of innovation and growth. *Harvard Business Review*, 89(1), 62-77.

Sommerville, I. (2011). *Software Engineering* (9th ed.). Addison-Wesley.

Turban, E., King, D., Lee, J. K., Liang, T. P., & Turban, D. C. (2018). *Electronic Commerce 2018: A Managerial and Social Networks Perspective*. Springer.